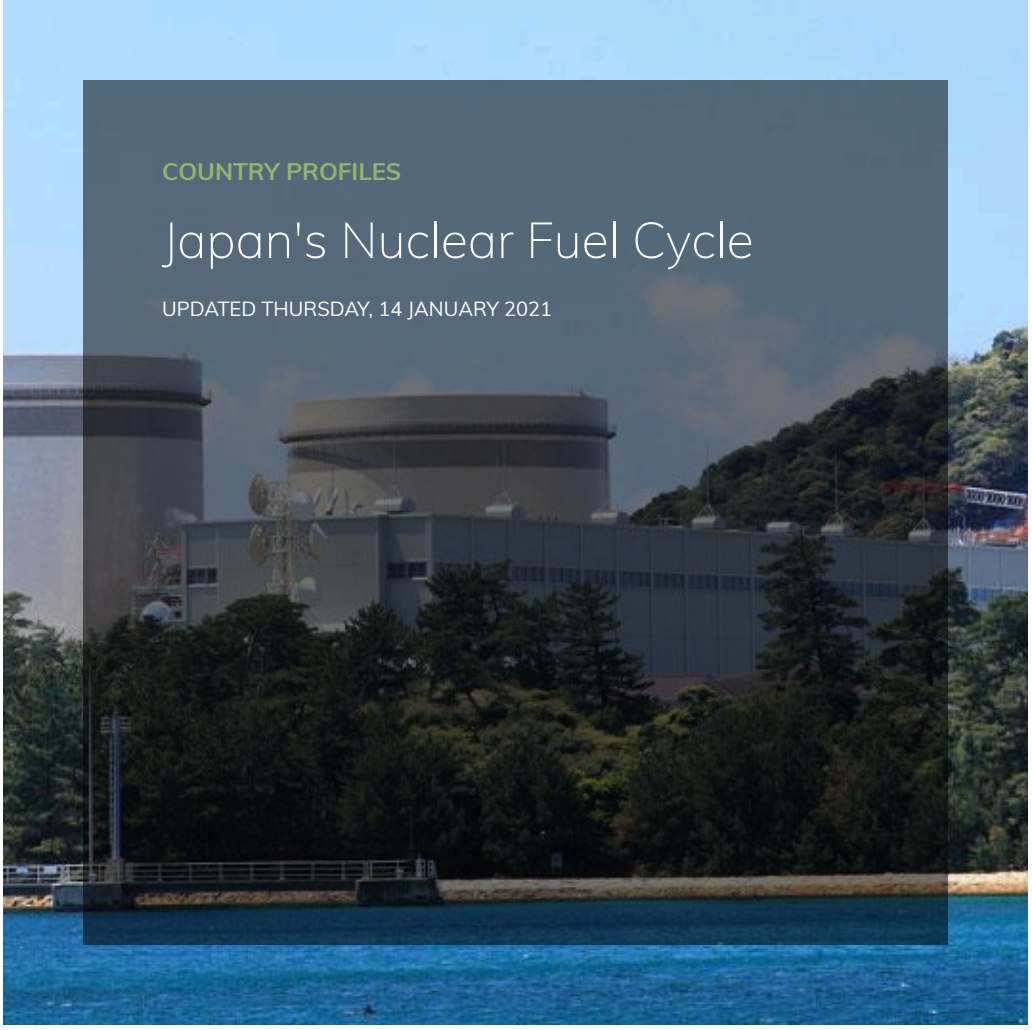

- Japan has a full fuel cycle set-up, including enrichment and reprocessing of used fuel for recycle.
- Japan has been a leading country in nuclear R&D, notably with fast reactors.
- Following the Fukushima accident its regulatory structure was completely overhauled.

Despite being the only country to have suffered the devastating effects of nuclear weapons in wartime, with over 100,000 deaths, Japan embraced the peaceful use of nuclear technology to provide

Privacy - Terms

a substantial portion of its electricity. See Nuclear Power in Japan paper.

Uranium supply

Japan has no indigenous uranium. Its annual requirements (of up to 8000 tU prior to the Fukushima accident) are normally met from Australia (about one-third), Canada, Kazakhstan and elsewhere.

Japanese companies have taken equity in overseas uranium projects.

In Kazakhstan, Itochu agreed to purchase 3000 tU from Kazatomprom over ten years in 2006, and in connection with this Japanese finance contributed to developing the West Mynkuduk deposit in Kazakhstan (giving Sumitomo 25%, Kansai 10%). In 2007 Japanese interests led by Marubeni and Tepco bought 40% of the Kharasan mine project in Kazakhstan and will take 2000 tU/yr of its production. A further agreement on uranium supply and Japanese help in upgrading the Ulba fuel fabrication plant was signed in May 2008. In March 2009 three Japanese companies – Kansai, Sumitomo and Nuclear Fuel Industries – signed an agreement with Kazatomprom on uranium processing for Kansai plants.

In Uzbekistan, a Japan-Uzbek intergovernmental agreement in September 2006 was aimed at financing Uzbek uranium development and in October 2007 Itochu Corporation agreed with Navoi Mining & Metallurgy Combinat (NMMC) to develop technology to mine and mill the black shales, particularly the Rudnoye deposit, and to take about 300 tU/yr from 2007. Then in February 2011 Itochu signed a 10-year "large-scale" uranium purchase agreement with NMMC.

In Australia, Mitsui joined Uranium One's Honeymoon mine project in 2008 as a 49% joint venture partner. Then early in 2009, a 20% share in Uranium One Inc was taken by three Japanese companies, giving overall 59% Japanese equity in Honeymoon. In July 2008 Mitsubishi agreed to buy 30% of West Australia's Kintyre project for US$ 495 million, with Cameco (70%). In February 2009 Mega Uranium sold 35% of the Lake Maitland project to the Itochu Corporation (10% of Japanese share) and Japan Australia Uranium Resources Development Co. Ltd. (JAURD), acting on behalf of Kansai Electric Power Company (50%), Kyushu Electric Power Company (25%) and Shikoku Electric Power Company (15%) for US$ 49 million.

In Namibia, Itochu Corporation bought a 15% stake in Kalahari Minerals, in March 2010, for US$ 92 million. Kalahari owns 41% of Extract Resources, which is developing the Husab project. Then in July 2010 Itochu bought a 10.3% direct stake in Extract for US$ 153 million, mostly from Polo Resources, giving it 16.43% overall in the project.

Fuel cycle – front end

Japan has been progressively developing a complete domestic nuclear fuel cycle industry, based on imported uranium.

Enrichment

The Japan Atomic Energy Agency (JAEA) operated a small uranium refining and conversion plant, as well as a small centrifuge enrichment demonstration plant, at Ningyo Toge, Okayama prefecture, but these facilities are now decommissioned.

While most enrichment services are still imported, Japan Nuclear Fuel Ltd (JNFL) operates a commercial enrichment plant at Rokkasho – RE2A. This began operation in 1997 using indigenous technology and had seven cascades each of 150,000 SWU/yr, though only one had been operating. The plant is being equipped with a new centrifuge design, and 37,500 SWU/yr came fully online in March 2012 after a 15-month break in operations. A further 37,500/yr SWU came online in May 2013, and in mid-2014, 1.05 million SWU was operational. The design capacity of 1.5 million SWU/yr is expected to be reached in about 2021. JNFL's main shareholders are the power utilities. In May 2017, Japan's Nuclear Regulation Authority (NRA) approved its operation in line with 2013 standards.

Japan had 6400 tonnes of uranium recovered from reprocessing (RepU) and stored in France and the UK, where the reprocessing was carried out. In 2007 it was agreed that Russia's Atomenergoprom would enrich this for the Japanese utilities who own it.

Fuel fabrication

At Tokai-mura, in Ibaraki prefecture north of Tokyo, Mistubishi Nuclear Fuel Co Ltd (MNF) operates a 440 tU/yr fuel fabrication

facility, which started up in 1972 and has had majority shareholding by Mitsubishi Materials Corporation (MMC). In April 2009 this was restructured as a comprehensive nuclear fuel fabrication company to supply Japanese customers with uranium fuel assemblies for pressurized water reactors (PWRs), boiling water reactors (BWRs) and high-temperature gas-cooled reactors (HTRs), as well as mixed oxide (MOX) fuel assemblies. It would also provide related services, including uranium reconversion from 2014. The new shareholdings are MHI 35%, MMC 30%, Areva 30% and Mitsubishi Corporation 5%, with capital of JPY 11.4 billion. In October 2009 it was announced that a new 600 t/yr plant using Areva's dry process technology would be built by the company. MMC has a plant at Okegawa, Saitama prefecture, to make zirconium alloy tubing, with a capacity of 800 tU/yr. As part of the new partnership, MHI and Areva in 2010 announced a 50-50 joint venture to manufacture APWR fuel at Areva's Richland plant in Washington state, but this has not proceeded.

At Kumatori and Tokai, Nuclear Fuel Industries (NFI) operates two fuel fabrication plants which have operated from 1976 and 1980 respectively. Kumatori (284 tU/yr) produces PWR and BWR fuel, Tokai (250 tU/yr capacity) is also set up to produce HTR and FNR fuel. NFI is also involved in a project to design MOX fuel for Areva to manufacture for Japanese power plants. In 2009 Westinghouse bought the 52% share of NFI owned by Furukawa and Sumitomo for $100 million.

Global Nuclear Fuel - Japan (GNF-J) is part of the GE-led Global Nuclear Fuel joint venture with Hitachi and Toshiba set up in 1970, which designs and manufactures fuel for BWRs. It has a 750 tU/yr plant at Yokosuka in Kanagawa prefecture. In March 2017 the NRA approved this as meeting 2013 safety standards and METI approved its restart.

JAEA has some experimental mixed oxide (MOX) fuel fabrication facilities at Tokai for both the Fugen ATR and the FBR program, with capacity about 10 t/yr for each. See also MOX section below.

Fuel Cycle – back end

For energy security reasons, and notwithstanding the low price of uranium for many years, Japanese policy since 1956 has been to maximise the utilisation of imported uranium, extracting an extra 25-30% of energy from nuclear fuel by recycling the unburned

uranium and plutonium as mixed-oxide fuel (MOX). The AEC reaffirmed this in 2005.

At the end of 2013 Japan's ten power companies were reported to have 55,610 used fuel assemblies, 13,236 tonnes, stored at 18 nuclear power plants. These occupied 55% of available pool storage space. In September 2016, FEPC reported that there were 14,830 tonnes of used fuel in wet and dry storage at Japan's power plants, filling about 71% of the existing storage capacity. This excludes about 3000 tonnes of used fuel stored at the Rokkasho reprocessing plant.

Tokai

Tokai is the main site of JAEA's R&D on HLW treatment and disposal.

At Tokai, JNC (now JAEA) operated a 90 t/yr pilot **reprocessing plant** using Purex technology which treated 1140 tonnes of used fuel between 1977 and its final batch early in 2009. Of this, 653 tU was from BWRs, 276 tU from PWRs and 111 tU from the 165 MWe Fugen prototype advanced thermal reactor. It processed 5401 used fuel assemblies, with a Pu-U mixed product as well as reprocessed uranium. Since 2006 when commercial contracts ran out the plant has focused on R&D, including reprocessing of MOX fuel from the Fugen ATR reactor. It was listed as having 40 t/yr MOX reprocessing capacity. In September 2014 JAEA announced that it would close down the whole facility rather than spend JPY 100 billion ($915 million) to upgrade it to new safety, especially seismic, standards. The Rokkasho plant would take over further work. In June 2017 it presented decommissioning plans to the Nuclear Regulation Authority (NRA), with work to take 60 years and cost JPY 987 billion. JAEA said the fuel pool at the reprocessing plant stores 265 spent fuel assemblies, equivalent to 40.7 tonnes of heavy metal, from JAEA's Fugen ATR.

JAEA also operated the associated pilot high-level waste (HLW) **vitrification facility** at Tokai from 1994 to 2007, producing 247 canisters of vitrified waste. This was restarted in January 2016 to vitrify 400 m^3 of liquid HLW, and shut down in June 2017 after vitrifying 48 tonnes of liquid HLW, filling 59 canisters, but leaving much more to do. Problems were reported as being similar to those encountered by JNFL at Rokkasho in 2008. In November 2017 the NRA approved restarting the plant in April 2019 to treat the remaining 379 m^3 of HLW by early 2029.

JAEA operates spent fuel storage facilities at Tokai containing 110 tonnes of fuel in 2014, and was proposing a further one.

Reprocessing in Europe

Until a full-scale plant was ready in Japan, the reprocessing of used fuel has been largely undertaken in Europe by BNFL and AREVA (4193 tU and 2944 tU respectively), with vitrified high-level wastes being returned to Japan for disposal. Areva's reprocessing finished in 2005, and commercial operation of JNFL's reprocessing plant at Rokkasho-mura was scheduled to start in 2008 (now 2021). Used fuel has been accumulating there since 1999 in anticipation of its full-scale operation (shipments to Europe finished in 1998). Japan at the end of 2012 had a total of 14,460 tonnes of used fuel in storage, mostly at reactors. In March 2017 JNFL had 2968 tonnes of used fuel in storage at Rokkasho, with a capacity of 3000 tonnes.

Reprocessing involves the conventional Purex process, but Toshiba is developing a hybrid technology using this as stage 1 to separate most uranium, followed by an electrometallurgical process to give two streams: actinides (plutonium and minor actinides) as fast reactor fuel, and fission products for disposal.

Government reviews

In April 2012 the government announced a full review of nuclear fuel cycle options, considering both economic and other criteria. The review committee started by considering technical options: one scenario involved direct disposal of used reactor fuel, two scenarios involved this being reprocessed and with fuel materials recycled as mixed-oxide fuel. Two more scenarios looked at the use of fast reactors and fast breeder reactors. A review of policy options based on these then followed. Finally, these were combined with the addition of a time axis with mid-to-long term scenarios. This review quantified the amount of plutonium and used fuel generated by each option as well as looking at broader impacts such as energy security, the international perspective, and the impacts of the changes resulting from each of the potential policies.

Meanwhile the AEC fuel cycle subcommittee has updated cost estimates for different used fuel options considering both 20% and 35% nuclear contributions to electricity in 2030. In each case reprocessing and recycle of used fuel is economically much better – by about 20% – than direct disposal.

In June 2012 the AEC brought all this together focused on three options to 2030, and sent them to the Energy & Environment Council (Enecan) along with a recommendation that any R&D on fast reactors should continue with international cooperation and as a means of waste treatment. Enecan in mid-September 2012 confirmed that reprocessing would continue. It was abolished later in the year, and METI's Advisory Committee for National Resources and Energy became responsible for such energy plans.

Spent Fuel Reprocessing Organisation from 2016

In May 2016 parliament passed a bill aimed at "taking measures necessary for the steady implementation of the reprocessing of used nuclear fuel" and MOX fuel fabrication, to go into effect within six months, amending a 2005 law on funding these activities. The bill creates a new entity responsible for reprocessing, the Spent Fuel Reprocessing Organisation (SFRO), which will collect funds and contract out reprocessing and MOX fuel fabrication to JNFL. It requires Japan's nuclear utilities to pay annual 'contributions' to the SFRO to cover the expected cost for reprocessing of all spent fuel they produce in the previous fiscal year, and for turning all of the resulting separated plutonium into MOX fuel. The contributions will be based on the amount of the electricity generated. This is in place of the former 'deposit system' which was restricted to the Rokkasho plant. The organisation was established in October 2016 in Aomori, and the ten nuclear generating companies and EPDC/J Power have notified the Nuclear Regulation Authority (NRA) of their assent to the new arrangements. In November 2016, the SFRO commissioned JNFL to continue reprocessing used fuel and to receive and manage HLW to be returned from France. The SFRO plans to assign MOX production to JNFL in 2019.

After consulting the AEC the METI minister must approve the organisation's plan from year to year and ensure that all used fuel is reprocessed and the plutonium separated for MOX. METI had noted that while reprocessing and the use of MOX fuel are key parts of the Basic Energy Plan approved by the cabinet in April 2014, they could be adversely impacted by impending full liberalisation of the electricity market, hence the new law.

Until the new system takes effect, Japan's ten power companies deposited fees for future reprocessing at Rokkasho with the Radioactive Waste Management Funding Research Centre (RWMC). The fee is JPY0.5 (0.4 US cents) per kilowatt-hour of nuclear electricity generated. This is supervised by METI's energy

arm, the Agency for Natural Resources and Energy (ANRE). ANRE reported that the fee deposits at RWMC amounted to JPY 2.4 trillion ($21 billion) as of March 2015.

Rokkasho complex – reprocessing and wastes

In 1984, the Federation of Electric Power Companies (FEPC) applied to the Rokkasho-mura village and Aomori prefecture for permission to construct a major complex including uranium enrichment plant, low-level waste (LLW) storage centre, HLW (used fuel) storage centre, a reprocessing plant, and a MOX fuel fabrication plant.

To undertake all these activities, Japan Nuclear Fuel Ltd (JNFL) was set up in 1992 as a joint stock company by the electric power utilities, with some wider shareholding. Pending commercial reprocessing, the ten power companies deposit fees for future reprocessing with the RWMC on the basis of kilowatt-hours generated, as described above. A new entity to take over from RWMC is proposed, along with a new basis for funding.

In 2015, 91% of JNFL is owned by ten power companies. Currently JNFL operates both LLW and HLW storage facilities at Rokkasho at the north end of Honshu (main island), near Higashidori nuclear power plant. Its 800 t/yr reprocessing plant is under construction and is being commissioned. The used fuel storage capacity is 20,400 tonnes.

In October 2004 the Atomic Energy Commission (JAEC) advisory group decided by a large majority (30 to 2) to proceed with the final commissioning and commercial operation of JNFL's 800 t/yr Rokkasho reprocessing plant (RRP), costing some JPY 2.4 trillion (US$ 20 billion). The Commission rejected the alternative of moving to direct disposal of spent fuel, as in the USA. This was seen as a major confirmation of the joint industry-government formulation of nuclear policy for the next several decades.* The cost had then increased to JPY 2.94 trillion ($25 billion), with 2021 operation anticipated.

* A 2004 government study showed that projected over the next 60 years it would be significantly more expensive to reprocess – at 1.6 yen/kWh, compared with 0.9 - 1.1 yen for direct disposal. This translates to 5.2 yen/kWh overall generating cost compared with 4.5 - 4.7 yen, without considering the implications of sunk investment in the new plant, or apparently the increased price of uranium since 2004.

In November 2011 the AEC released results of a further study on the same matter. At the reference 3% discount rate, direct disposal after interim storage would cost about JPY 1 per kWh, while immediate reprocessing of all Japanese spent fuel would cost JPY 1.98 per kWh. Storage for 20 years followed by reprocessing would cost JPY 1.39 per kWh. JNFL policy from May 2018 is to store used fuel for 15 years before reprocessing.

The Rokkasho reprocessing plant was due to start commercial operation in November 2008, following a 28 month test phase plus some delay at the end of 13 years construction. The test phase treated 197 t in PWR fuel and 134 t in BWR fuel in four cycles to January 2008. Based on previous figures, this would have yielded about 1.8 tonnes of fissile plutonium (in reactor-grade material). The intended start date was then moved to October 2013, the five-year delay being due to problems in the locally-designed vitrification plant for HLW at the end of the line (see below). The main plant is based on Areva's La Hague technology, and in late 2007 the twenty-year cooperation agreement with Areva was extended and related specifically to Global Nuclear Energy partnership (GNEP, now IFNEC) goals. The modified PUREX process now employed leaves some uranium with the plutonium product – it is a 50:50 oxide mix recovered through a mixed denitration process, so there is no separated plutonium at any time, alleviating concerns about potential misuse.

Active testing at the new vitrification plant attached to the Rokkasho reprocessing plant commenced in November 2007, with separated high-level wastes being combined with borosilicate glass. The plant takes wastes after uranium and plutonium are recovered from used fuel for recycle, leaving 3% of the used fuel including minor actinides as high-level radioactive waste. However, the furnaces (developed at Tokai, rather than being part of the French technology) proved unable to cope with impurities in the wastes, and commissioning was repeatedly delayed. The vitrification plant was shut down in October 2008 and in 2010 JNFL decided to redesign the unit to better control temperature of the molten glass. One of the two rebuilt furnaces was tested successfully in July 2012, producing ten batches of vitrified HLW, and a second test run successfully produced 33 canisters by mid-January 2013, confirming its 70 litres/hour HLW rate. The other system was tested similarly by May 2013, producing 25 logs of vitrified HLW, each from 70 litres of liquid HLW.

JNFL was ready to commission the reprocessing plant in October 2013, but had to await Nuclear Regulation Authority (NRA) inspections, following new NRA fuel cycle facility regulations

published in November 2013. The NRA found no problem with the plant in respect to earthquake and tsunami issues – it approved a seismic increase to 700 Gal, but its overall safety review took much longer than envisaged. In addition, a report on reprocessing commissioned by METI's Agency for Natural Resources and Energy (ANRE) in mid-2015 was delivered to the Atomic Energy Commission (JAEC) in January 2016. It considered the role of the new Spent Fuel Reprocessing Organisation (see subsection above), albeit using the services of JNFL. In July 2017 JNFL said that it needed to spend an extra JPY 700 billion to meet NRA requirements, bringing the total plant cost to JPY 2.94 trillion ($26 billion). JNFL expected to complete the work in 2018, but has since deferred the start-up date to 2021.

The total reprocessed in the active test phase to March 2015 was 219 tU in BWR fuel and 206 tU in PWR fuel, yielding 364 tU as uranium oxide and 6.7 tU+Pu as well as 3.6 tPu in MOX powder (2.3 t fissile Pu). The vitrified waste was in 346 canisters. However, in November 2014 and again in November 2015 JNFL announced that due to the slow NRA safety checks, the plant would not start commercial operation until September 2018. In its first six months JNFL plans to reprocess 282 BWR fuel assemblies (48 t fuel) and 73 PWR assemblies (128 t fuel). In FY 2019 (from April 2019) it plans to reprocess 1129 BWR assemblies (192 t fuel) and 291 PWR ones (128 t) – these being the same amounts earlier intended for FY 2015.

The new Rokkasho plant will treat 14,000 tonnes of used fuel stockpiled to end of 2005 plus 18,000 tonnes of used fuel arising from 2006, over some 40 years. Eventually at 800 t/yr it will produce about 4 tonnes of fissile plutonium per year, enough for about 80 tonnes of MOX fuel. It is not planned to treat fast reactor fuel there.

The Rokkasho facility has storage capacity for 2880 canisters of vitrified HLW. In April 2015 it had 1574 canisters, 1310 of these from La Hague and 264 from Sellafield. In October 2016 it accepted the last batch of used fuel from utilities so that it had 2968 tonnes of used fuel stored, its capacity being 3000 tonnes.

In December 2013 new NRA regulations came into effect, including seismic design basis of 600 Gal (up from 375 Gal). JNFL filed requests in January 2014 for safety reviews of the reprocessing plant, the J-MOX fuel fabrication plant (annual capacity of 140 tonnes), a uranium enrichment plant (1,050 SWU/yr capacity), and for a waste storage facility – all at Rokkasho. In February 2016 the NRA approved an increase to 700 Gal for the reprocessing plant,

for which JNFL had hoped to get an operating licence by September 2018.

The NRA approved safety checks in May 2020. The plant is now expected to start operation in 2022.

The 2013 NRA regulations specified that interim storage of spent fuel should be in dry storage with convection cooling. This applies principally to Mutsu.

Mutsu storage

In 2010 Recyclable-Fuel Storage Co (Tepco 80%, Japco 20%) obtained approval to construct a facility at Mutsu in the northern Aomori prefecture to store used fuel from Tepco and Japco nuclear plants for some 50 years before reprocessing at the Japan Nuclear Fuel plant in that region. Initial capacity is 3000 tonnes, in dry casks, and a further stage after 10-15 years will add 2000 t capacity. NISA approved this in August 2010. Construction was completed in October 2013, but it awaited full NRA review according to the new regulations including earthquake and tsunami countermeasures. In August 2017 the NRA approved a Ss level of 620 Gal for it. Following full approval in September 2020, it is expected to come into service in fiscal year 2021. About 70% of the JPY 100 billion cost (2006 estimate) is reported to be the casks.

Mixed-oxide fuel

All Japanese nuclear reactor operators will use will use plutonium as mixed oxide (MOX) fuel. This was planned to be in 16-18 reactors from 2015 under the 'pluthermal' programme, but the target date is now indefinite. About 6 tonnes of fissile plutonium per year (in about 9 tonnes of reactor-grade Pu) was expected to be loaded into power reactors, slightly more than anticipated annual arisings from reprocessing. Meanwhile MOX fuel fabricated in Europe from some 40 tonnes of separated reactor-grade plutonium from Japanese spent fuel can be used. However, local concerns about MOX fuel use has slowed implementation of the 1994 'pluthermal' programme, and not until late 2009 was there a commercial Japanese reactor running with MOX.

By end of 2010 the Nuclear & Industrial Safety Agency (NISA) on behalf of the Ministry (METI) had approved the use of MOX fuel in ten reactors, including: Takahama 3&4, Fukishima I-3, Kashiwazaki-Kariwa 3, Genkai 3, Hamaoka 4, Onagawa 3, Ikata 3 and Shimane

2. This was expected to occur progressively to 2012, after modifications to the reactors to take a one-quarter or one-third core of MOX. NRA permission for MOX use in Tomari 3 was pending.

In 2008 the Shizuoka prefecture accepted Chubu's plans to use MOX in its Hamaoka 4 plant. Fukui prefecture accepted Kansai's planned use of MOX at Takahama 3 and 4 from 2010. Hokkaido accepted Hokkaido Electric Power's use of MOX at Tomari 3, and in 2010 Fukushima prefecture agreed to MOX use in TEPCO's Fukushima I-3 reactor, making a total of 12 reactors allowed to use it. In the event only Takahama 3, Fukishima I-3, Genkai 3, and Ikata 3 were using it in 2011.

So far, Japan has received five shipments containing over two tonnes of its (reactor-grade) plutonium from Europe. The first shipment, in 1992, was simply plutonium oxide and earmarked for use in the Monju prototype FBR.

Subsequent shipments have been in the form of MOX fuel for light water reactors. The first MOX shipment was in 1999. Part of this shipment from BNFL and intended for use in Kansai Electric Power Co's Takahama plant was found to contain falsified quality control data, so that material was returned to the UK in 2002. The balance was for Tepco's Fukishima I-3. The second MOX shipment in 2001 consisted of fuel from BNFL for use in TEPCO's Kashiwazaki-Kariwa 3 reactor. The third MOX shipment was fuel for Chubu's Hamaoka BWR, Shikoku's Ikata PWR and Kyushu's Genkai PWR, and arrived from France in May 2009. The fourth MOX shipment in 2010 from France contained 12 assemblies for Kansai's Takahama 4 and 20 for the second load at Genkai 3. The fifth shipment which arrived in June 2013 contained 20 assemblies for Kansai's Takahama 3 plant. The sixth shipment in 2017 was for 16 assemblies for Takahama 4.

In November 2009 Kyushu Electric Power started using MOX in its Genkai 3 reactor. During a scheduled refuelling outage the company replaced about one-third of the 193 PWR fuel assemblies, 16 of them comprising MOX fuel. Shikoku Electric Power Co started Ikata 3 with some MOX fuel in March 2010, and Tepco started up Fukishima-Daiichi 3 BWR with MOX fuel in September 2010. Kansai started using MOX in its Takahama 3 PWR in January 2011, but in mid 2011 deferred its use in unit 4. It plans to use it in both Ohi reactors. Chubu postponed MOX use in Hamaoka 4. Hokkaido plans to load MOX into Tomari 3 and Shika 1 by 2016. Tohoku plans to use MOX in Onagawa 3. In 2016 Areva in France started making 16 MOX fuel assemblies for Takahama 4, and in August 2017 Areva New NP signed an agreement with

Nuclear Fuel Industries to supply 32 MOX assemblies for Kansai's Takahama 3&4. These will be made at Melox in France.

In March 2018 the Federation of Electric Power Companies (FEPC) reiterated to the Japan Atomic Energy Commission (JAEC) its members' intention to use plutonium as MOX fuel in 16-18 nuclear units. FEPC said it would be more specific when the Rokkasho reprocessing plant was closer to operation in 2021.

For its new Ohma ABWR plant, designed to run on a full MOX core, J-Power has signed a contract with Areva to supply the first three years' fuel, fabricated from Japanese plutonium separated in France. Areva also has MOX fabrication contracts with Chubu, Kyushu, Shikoku and Kansai.

At the end of 2018 there was 9 tonnes of separated reactor-grade plutonium (about 66% fissile) stored domestically, plus a total of 36.6 t in the UK and France. That total of 45.6 t was a modest decrease from an estimated 47.3 t at the end of 2017.

J-MOX plant

In April 2005 the Aomori prefecture approved construction of the JNFL's J-MOX plant at Rokkasho, adjacent to the reprocessing plant. The Governor urged the Federation of Electric Power Companies "to step up their efforts towards realisation of the MOX-use program." The approval was seen as a significant step forward in closing the fuel cycle in Japan, and was strongly supported by the federal government, JAEC, and utilities.

JNFL applied for two of the four licences needed to build and operate the 140 t/yr plant. Construction of the plant started in October 2010 after a three-year delay due to revision of seismic criteria, which have since become 700 Gal. Operation of J-MOX is now expected from September 2022, and the cost has escalated to JPY 210 billion (US$ 2.05 billion). It will produce MOX with 4-9% plutonium.

In November 2006 Shikoku Electric Power contracted with Mitsubishi to manufacture 21 MOX fuel assemblies for its Ikata nuclear plant using 600 kg of reactor-grade plutonium. The plutonium had been recovered by Areva at La Hague from Shikoku's used fuel and the MOX was fabricated at Areva's Melox plant in France and shipped to Japan in March 2009.

With the delay in construction of the J-MOX plant, several other utilities have sought MOX fuel supplies from Areva in France.

Once MOX fuel is fully in routine use in Japan, it is expected that the Japanese stockpile of separated plutonium in Europe will be used up in about 15 years, with demand being about 6 tonnes per year of fissile plutonium (Puf) and output from Rokkasho only 4 tonnes Puf.

METI approved construction a used fuel storage facility for Tepco and Japco in Mutsu, at the same time as approving J-MOX. Government approval for both followed.

High-level waste

In 1995, Japan's first high-level radioactive waste (HLW) interim storage facility opened in Rokkasho-mura – the Vitrified Waste Storage Centre. The first shipment of vitrified HLW from Europe (from the reprocessing of Japanese fuel) also arrived in that year. The last of twelve shipments from France was in 2007, making a total of 1310 canisters. Shipments from UK started in 2010, with 1850 canisters to go in about 11 shipments. These include an equivalent amount of HLW to avoid the need to transport greater amounts of low-level waste (LLW). The first shipment from UK arrived in March 2010, the fourth in April 2014.

In 2005 Tepco and JAPC announced that a Recyclable Fuel Storage Centre would be established in Mutsu, with 5000 t capacity, to provide interim storage for up to 50 years before used fuel is reprocessed. See fuller description above.

Japan’s R&D program for HLW started in 1976, and progress reports on technical feasibility were published in 1992 and 2000. In May 2000, the Japanese parliament (the Diet) passed the Law on Final Disposal of Specified Radioactive Waste (the 'Final Disposal Law') which mandates deep geological disposal of high-level waste (defined as only vitrified waste from reprocessing spent reactor fuel). In line with this, the Nuclear Waste Management Organisation (NUMO) was set up in October 2000 by the private sector to implement a project for geological disposal of HLW. NUMO initiated the siting process with open solicitation of volunteer host municipalities for exploring the feasibility of constructing a final repository. This open solicitation approach was announced in December 2002 and information packages were sent to all municipalities in Japan. The Final Disposal Law was revised in 2007 to include transuranic wastes for geological disposal: some types of long-lived, low heat-generating waste from fuel reprocessing and other processes in the nuclear fuel cycle.* NUMO was responsible

for making plans for disposal, including site selection, demonstration of technology there, licensing, construction, operation, monitored retrievable storage for 50 years and closure of the repository. Some 40,000 canisters of vitrified HLW are envisaged by 2020, needing disposal – all the arisings from the Japanese nuclear plants until then.

* The supplementary waste disposal bill which was passed in 2007 said that final disposal is the most important issue in steadily carrying out nuclear policy. It called for the government to take the initiative in helping the public nationally to understand the matter by promoting safety and regional development, in order to get the final disposal site chosen with certainty and without delay. It also called for improvement in disposal technology in cooperation with other countries, revising the safety regulations as necessary, and making efforts to recover public trust by, for example, establishing a more effective inspection system to prevent the recurrence of data falsifications and cover-ups.

NUMO's open solicitation process aimed to shortlist those sites that were proffered and potentially suitable. The promising ones would be subject to detailed investigation from 2012. A third phase to 2030 would end with site selection. However, the government decided to become more involved in the process, and in October 2013 METI's Agency for Natural Resources and Energy (ANRE) established a working group to re-evaluate geological disposal technology and developments since 2000, taking account of NUMO's work. It reported in April 2014 that potential repository sites were available across Japan. The minister said in 2013: "The government will play an active role in choosing a permanent place. We will abandon the current system of waiting for volunteers to raise their hands."

In April 2014, HLW disposal was part of the new Basic Energy Plan, which included facilitating construction and use of new intermediate and dry storage facilities. Then in May 2015 the cabinet endorsed the proactive approach. Once possible locations are short-listed with AEC oversight, the government will seek local government consent to pursue plans for a deep geological repository. In January 2016 ANRE invited opinions from experts on an interim report concerning specific requirements and standards for "scientifically promising sites" for the final disposal of HLW. Based on two calls for public comment in August 2016 and March 2017, as well as other information, the Geological Disposal Working Group under METI's Advisory Committee for Natural Resources and Energy presented a report in April titled *Summary of Requirements*

and Criteria for a Nationwide Map of Scientific Features for Geological Disposal.

Then in July 2017 METI published on the NUMO website a scientific characteristic map based on these identified requirements and criteria. The map identifies regions that are likely to meet the necessary geological requirements for hosting a repository and could be included in a future detailed site selection survey. The map shows areas not suitable for a repository due to their proximity to volcanoes or active faults. Areas which have mineral resources are also excluded due to the potential of future drilling activities. Fukushima prefecture is also excluded to avoid any further "burden", as is Aomori prefecture hosting Rokkasho, due to a 1995 agreement. This leaves about two-thirds of the country as potentially suitable.

In November 2020 Japan's trade and industry minister approved investigation of potential sites in Suttu and Kamoenai. The two sites are south and north of Hokkaido Electric Power's Tomari reactors.

The ¥3700 billion ($33 billion) cost of the repository will be met by funds accumulated at ¥0.2/kWh from electricity utilities (and hence their customers) and paid to NUMO. By 2015 ¥1 trillion had been collected thus. This sum excludes any financial compensation paid by the government to local communities.

The technical aspects of Japan's HLW disposal concept is based on several decades' work under JAEA involving generic evaluation of repository requirements in Japan's geology. Since 2000 the Horonobe Underground Research Centre on Hokkaido has been investigating sedimentary rocks about 500m deep, and in November 2005 construction of the underground shafts and a 760-metre gallery was launched. This is a seismically stable area of Japan. JAEA runs the Tona Geoscience Centre at Toki, in Gifu prefecture, and has built a similar facility, the Mizunami Underground Research Laboratory (MIU) also in Gifu prefecture, in igneous rock about 1000m deep.

The basic repository concept involves sealing about 20 HLW canisters in a massive steel cask or overpack and surrounding this by bentonite clay. NUMO has built design options on this including those allowing inspection and retrieval over long periods. In particular the Cavern Retrievable (CARE) concept has emerged, involving two distinct stages: ventilated underground caverns with the wastes in overpacks (hence shielded) fully accessible, followed by backfilling and sealing the caverns after 300 years or so. The initial institutional control period allows radiological decay of the

wastes so that thermal load is much reduced by stage 2 and hence the concept allows a much higher density of wastes than other disposal concepts.

The CARE concept can be adapted for spent fuel, the cask then being similar to shipping casks for such except that a layer of shielding required due to higher thermal and radiation output could be removable before the cavern is backfilled and sealed. However, for spent fuel retrieval would be likely rather than merely possible, since it represents a significant potential fuel resource (via reprocessing), whereas vitrified HLW does not. Also spent fuel would require ease of access due to the need for safeguards inspections. Eventual backfill could include depleted uranium if that is then considered a waste.

In 2004 METI estimated the costs of reprocessing spent fuel, recycling its fissile material and management of all wastes over 80 years from 2005. METI's Electricity Industry Committee undertook the study, focused on reprocessing and MOX fuel fabrication including the decommissioning of those facilities (but excluding decommissioning of power reactors). Total costs over 80 years amount to some JPY 19 trillion, contributing almost one yen (US 0.9 cents) per kilowatt-hour at 3% discount rate. About one third of these costs would still be incurred in a once-through fuel cycle, along with increased high-level waste disposal costs and increased uranium fuel supply costs. Japan's policy however is based on energy security rather than purely economic criteria.

Funding arrangements for HLW were changed in October 2005 under the new Back-end Law which set up the Radioactive Waste Management Funding and Research Centre (RWMC) as the independent funds management body. All reserves held by utilities were to be transferred to it and companies then refunded as required for reprocessing.

Low- and Intermediate-level waste

JNFL operates a large LLW storage facility at Rokkasho. METI, with JNFL and FEPC, is seeking permission from the Aomori prefecture to build further low-level waste storage capacity there, adjacent to the reprocessing plant. In particular this will be for LLW and what is internationally designated as ILW returned from France from 2013.

JNFL announced in May 2018 that it planned a new facility at Rokkasho for the disposal of 42,240 cubic metres of LLW, including metals and plastics. Construction is to begin in April 2020, with

commissioning in 2023. It will be adjacent to the two present LLW storage plants which can hold 40,000 m^3 each – one for liquid and ash, the other for metals and plastics. Early in 2018 these were 75% full.

Tests are under way at Rokkasho regarding disposal of intermediate-level waste.

Decommissioning

The Japan Power Demonstration Reactor (JPDR) decommissioning program, following its closure in 1976, established techniques for the decommissioning of commercial power reactors by the Japan Atomic Energy Research Institute (JAERI). Phase I of the program started in 1981 to develop a set of techniques and Phase II was actual dismantling of JPDR over 1986-92.

The original Tokai 1 power station, a British Magnox reactor which started up at the end of 1965 and closed down in March 1998, is being decommissioned over 20 years, the first ten as "safe storage" to allow radioactivity to decay. Phase 1 (to 2006) comprised preliminary work, in Phase 2 (to 2011) the steam generators and turbines are being removed, and in Phase 3 (to 2018) the reactor will be dismantled, the buildings demolished and the site left ready for re-use. All radioactive wastes will be classified as low-level (LLW), albeit in three categories, and will be buried – the 1% of level I wastes 50-100 metres deep. The total cost is expected to be JPY 93 billion – JPY 35 billion for dismantling and JPY 58 billion for waste treatment including the graphite moderator (which escalates the cost significantly).

Fugen ATR (148 MWe, started up in 1978) closed in March 2003, and JAEA plans to decommission it and demolish to clear the site by 2029, at a total cost of about JPY 70 billion, including waste treatment and disposal. Plans for this were approved in February 2008.

Chubu's Hamaoka 1&2, earlier closed for safety-related upgrades, remained shut down following the 2007 earthquake, were written off, and are now being decommissioned. However, the company has not found a storage site for the wastes, so they remain on site. It expects to have about 20,000 tonnes of wastes by 2036 when the work is due to be completed.

In March 2011 units 1-4 of the Fukushima Daiichi plant (2719 MWe net) were seriously damaged in a major accident, and are written

off to be decommissioned. Units 5&6 were basically undamaged, but are written off from January 2014 to appease public opinion. Tepco established an internal entity, the Fukushima Daiichi Decontamination & Decommissioning Engineering Company, to focus on measures for decommissioning units 1-6 and dealing with contaminated water. The company commenced operations in April 2014.

In March 2015, Kansai announced that Mihama 1&2 PWRs would be retired, and Japan Atomic Power Co (Japco) said it would decommission its Tsuruga 1 BWR, all in Fukui prefecture. Then Chugoku Electric Power Co announced decommissioning of its Shimane 1 BWR in Shimane prefecture, and Kyushu Electric Power Co did the same for its Genkai 1 PWR in Saga prefecture. Some JPY 176.5 billion ($1.47 billion) has been set aside by the four utilities for decommissioning (Mihama 1&2: JPY 68 billion, Tsuruga JPY 36 billion).

Kansai and JAPC submitted plans for decommissioning to the Nuclear Regulation Authority (NRA) in February 2016 for approval. The amount of wastes arising is estimated in each application. JAPC estimates about 40 tonnes of high-level waste, 1990 tonnes of intermediate-level waste and 10,760 tonnes of low-level waste being included in the Tsuruga total. In April 2015 the All Japan Council of Local Governments with Atomic Power Stations (Zengenkyo) submitted a written request to METI, asking the national government to work responsibly on the decommissioning of nuclear power plants and to develop appropriate regulatory standards.

Kyushu submitted plans for decommissioning in December 2015, and it plans to decommission and dismantle Genkai 1 over 28 years, to 2043. To 2021 used fuel will be unloaded and some transferred to the Rokkasho reprocessing plant by 2029. Then the secondary steam supply system will be dismantled, followed by the main reactor vessel and internals and the steam generators all by 2036. The containment and buildings will be demolished over 2037 to 2043. Some 7000 tonnes of highly radioactive materials will result, along with 800 t low-level wastes and 2000 t of concrete and other very low-level wastes. In March 2020 the NRA approved decommissioning plans for Genkai 2.

Chugoku plans to decommission and dismantle Shimane 1 over 30 years to 2045, and in May 2016 filed an application with the NRA to this end. The reactor itself will be dismantled from 2030.

The NRA approved decommissioning plans for all five reactors in April 2017. Estimated costs for Mihama 1&2 are JPY 32.3 billion and JPY 35.7 billion; for Tsuruga 1, JPY 36.3 billion; for Shimane 1, JPY 38.2 billion; and for Genkai 1, JPY 36.4 billion.

Shikoku decided to retire Ikata 1 in March 2016 and the NRA approved the decommissioning plan in June 2017. Estimated cost is JPY 40.7 billion. In March 2018 it decided to decommission Ikata 2.

Tohoku announced in October 2018 that Onagawa 1 would be decommissioned, at an estimated cost of JPY 43.2 billion, with 69% of that already provided.

A plan for decommissioning JAEA's Monju prototype fast breeder reactor (FBR) has been officially adopted by the government-appointed team to consider the task, and it has approved JAEA's outline plan to carry it out over some 30 years. The government's policy calls for used fuel to be removed from the reactor and placed in an onsite storage pool within six years. This fuel – together with sodium coolant and other radioactive waste – is to then be removed from Fukui prefecture and reprocessed. Reprocessing may be in France, since the Rokkasho plant is not equipped for MOX from fast reactors. The government estimates that decommissioning will cost more than JPY 375 billion ($3.2 billion), including JPY 225 billion for maintenance, JPY 135 billion for dismantling and an initial JPY 15 billion for defuelling and preparations.

Japanese reactors being decommissioned

Reactor Name	Model	Reactor Type	Reference Unit Power (MWe)	Permanent Shutdown
Japan Power Demonstration Reactor (JPDR)	BWR-1	BWR	12	1976-03
Tokai 1	MAGNOX	GCR	137	1998-03
Fugen ATR	ATR	HWLWR	148	2003-03
Hamaoka 1	BWR-4	BWR	516	2009-01
Hamaoka 2	BWR-4	BWR	814	2009-01
Fukushima Daiichi 3	BWR-4	BWR	760	2011-05
Fukushima Daiichi 2	BWR-4	BWR	760	2011-05
Fukushima Daiichi 1	BWR-3	BWR	439	2011-05
Fukushima Daiichi 4	BWR-4	BWR	760	2011-05

Reactor Name	Model	Reactor Type	Reference Unit Power (MWe)	Permanent Shutdown
Fukushima Daiichi 6	BWR-5	BWR	1067	2013-12
Fukushima Daiichi 5	BWR-4	BWR	760	2013-12
Mihama 2	M (2-loop)	PWR	470	2015-04
Genkai 1	M (2-loop)	PWR	529	2015-04
Tsuruga 1	BWR-2	BWR	340	2015-04
Mihama 1	W (2-loop)	PWR	320	2015-04
Shimane 1	BWR-3	BWR	439	2015-04
Ikata 1	M (2-loop)	PWR	538	2016-05
Monju	-	FBR	246	2017-12
Ohi 1	W (4-loop)	PWR	1120	2018-03
Ohi 2	W (4-loop)	PWR	1120	2018-03
Ikata 2	M (2-loop)	PWR	538	2018-05
Onagawa 1	BWR-4	BWR	498	2018-12
Genkai 2	M (2-loop)	PWR	529	2019-04
Fukushima Daini 4	BWR-5	BWR	1067	2019-09
Fukushima Daini 1	BWR-5	BWR	1067	2019-09
Fukushima Daini 2	BWR-5	BWR	1067	2019-09
Fukushima Daini 3	BWR-5	BWR	1067	2019-09

JAEA has been responsible for research on reactor decommissioning. However, in August 2013 an International Research Institute for Nuclear Decommissioning (IRID) was set up in Japan by JAEA, Japanese utilities and reactor vendors, with a focus on Fukushima 1-4.

JAEA in June 2017 submitted its decommissioning plan for the Tokai reprocessing plant to the NRA, which was approved in June 2018.

In August 2014 the Nuclear Damage Compensation and Decommissioning Facilitation Corporation (NDF) was set up by government as a planning body with management support for R&D projects, taking over IRID's planning role. It will work closely with IRID, whose focus then became developing mid- and long-term decommissioning technologies, though early in 2015 it presented as a private consortium of companies bidding for Japanese government national research projects. NDF will also work closely with Tepco Fukushima Daiichi D&D Engineering Co. which has responsibility for operating the actual decommissioning work there. The NDF will be the main body interacting with government (METI) to implement policy.

Research & development

The Japan Atomic Energy Research Institute (JAERI) and the Atomic Fuel Corporation were set up in 1956. The latter was renamed PNC in 1967 and reconstituted as Japan Nuclear Cycle Development Institute (JNC) in 1998. A merger of JNC and JAERI in 2005 created the Japan Atomic Energy Agency (JAEA) under the Ministry of Education, Culture, Sports, Science & Technology (MEXT). JAEA is now a major integrated nuclear R&D organization, with 4400 employees at ten facilities and annual budget of ¥161 billion (US$ 1.7 billion).

JAEA runs a number of research reactors including JRR-3, a 20 MW unit supporting neutron beam experiments, and JRR-4, a 3.5 MW reactor used for medical irradiation, activation analysis and training. The Nuclear Safety Research Reactor (NSRR) is used for that purpose.

JAEA's Japan Materials Testing Reactor (JMTR) at the Oarai R&D Centre was refurbished for 2011 resumption of operation, to produce some radioisotopes, notably Mo-99, as well as enable basic research on LWR fuel and materials, and other applications. The JMTR was initially converted from 93% HEU fuel to 45% enriched fuel in 1991, and then to 19.8% enriched fuel in 1994.

High temperature reactors

At the end of 1998 JAERI's (now JAEA's) small prototype gas cooled reactor, the 30 MWt High Temperature Engineering Test Reactor (HTTR) started up at the Oarai Nuclear Hydrogen and Heat Application Research Centre near Tokyo. This was Japan's first

graphite-moderated and helium-cooled reactor. It runs at 850°C and in 2004 achieved 950°C, then demonstrated stable heat at this level over 50 days in 2010. This will allow its application to chemical processes such as thermochemical production of hydrogen. Its fuel is ceramic-coated particles with low-enriched (average 6%) uranium incorporated into hexagonal graphite prisms, giving it a high level of inherent safety. It is designed to establish a basis for the commercialisation of second-generation helium-cooled plants running at high temperatures for either industrial applications or to drive direct cycle gas turbines. Fuel burn-up of 90 GWd/t has been established, with the help of tests at the WWR-K research reactor in Kazakhstan. Fuel with target burn-up of 160 GWd/t is being developed (100 GWd/t was previously achieved in Germany and the USA).

By 2015 an iodine-sulfur (IS) plant producing 1000 m^3/h of hydrogen was expected to be linked to the HTTR to confirm the performance of an integrated production system. Silicon carbide ceramic components have been developed to survive the high-temperature acidic environment. However, HTTR was shut down in February 2011 for planned inspections and remained closed pending new post-Fukushima regulations which were announced in July 2013. In May 2020 the NRA approved modifications, and JAEA expected to restart it as soon as possible. JAEA then plans to connect a helium gas turbine and hydrogen production system and to prove the operation of these through to 2030 – the HTTR-GT/H_2 plant. This is designed to produce 1 MWe of electricity plus 30m^3/h of hydrogen from 0.7 MWt.

Based on the HTTR, JAEA has designed a 600 MWt high temperature reactor (HTR) called the GTHTR300C using Brayton cycle at 850-950°C to produce up to 300 MWe. Fuel is prismatic type TRISO with 14% uranium enrichment. Average burn-up is 120 GWd/t with 48-month fuel cycle. Up to four modules would comprise a power plant. This is for thermochemical hydrogen production at 120 t/day, or alternatively to produce 300 MWe and use the waste heat in multi-stage flash (MSF) desalination, the projected water cost for 55,000 m^3/day being half that of using gas-fired CCGT. It is being developed with Mitsubishi Heavy Industries (MHI), Toshiba/IHI and Fuji, and target for commercialisation is about 2030.

JAEA's small HTR50S reactor based on the HTTR is a conceptual design for industrial process and heat and/or power generation. This is 50 MWt with dual reactor outlet temperatures of 750°C and 900°C with maximum use of conventional technologies in order to deploy them in developing countries in the 2020s.

Initially this would use steam cycle for power generation, then improve the fuel, and then Increase the reactor outlet temperature to 900°C and install an intermediate heat exchanger to demonstrate helium turbine and hydrogen production using the IS process.

Early in 2019 the Japan Atomic Energy Agency (JAEA) formed a joint venture with Penultimate Power UK to build a 10 MWe SMR there based on the HTTR, for power and process heat. Plans include scaling up the design to 100 MWe and building a factory in the UK for multiple plants.

MHI and JAEA have proposed a modular HTR for power generation using steam cycle, with 50-100 MWe output (120 or 250 MWt), and similar fuel arrangement to HTTR. MHI has started the conceptual design of the MHR-100GT.

In 2014 the government included HTR research in its draft basic energy plan, and in May the Nuclear Science and Technology Committee of MEXT established a working group to evaluate the current R&D situation of HTRs and discuss their future direction, based on domestic and foreign needs. In July 2018 the government confirmed HTRs as part of its energy policy, and in June 2019 the Cabinet affirmed HTRs for thermochemical hydrogen production.

JAEA has a major agreement with Kazakhstan's National Nuclear Centre relating to the design, construction and operation of an HTR of about 50 MW at Kurchatov city. See also Kazakhstan information page. It also has HTR research agreements with South Korea and China.

JAEA is proposing to develop a 100 MW demonstration HTR at Abu Dhabi in the UAE, with Emirates Nuclear Energy Corp (ENEC), and expedited by the Japan Engineers Federation (JEF).

Under a 2007 cooperation agreement with Indonesia's National Atomic Energy Agency (BATAN), JAEA in 2014 agreed to include research and development of HTRs. Prior to the introduction of commercial reactors in Indonesia, BATAN plans to build a test and demonstration HTR of about 3-10 MWe, though this is contracted to Russia.

Reduced-moderation water reactor

The reduced-moderation water reactor (RMWR) being developed in Japan is a light water reactor, essentially as used today, with the fuel packed in more tightly to reduce the moderating effect of the

water. Considering the BWR variant (resource-renewable BWR – RBWR), only the fuel assemblies and control rods are different. In particular, the fuel assemblies are much shorter, so that they can still be cooled adequately. Ideally they are hexagonal, with Y-shaped control rods. The reduced moderation means that more fissile plutonium is produced and the breeding ratio is around 1 (instead of about 0.6), and much more of the U-238 is converted to Pu-239 and then burned than in a conventional reactor. Burn-up is about 45 GWd/t, with a long cycle. Initial seed (all??) MOX fuel needs to have about 10% Pu. The void reactivity is negative, as in conventional LWR. A Hitachi RBWR design based on the ABWR-II has the central part of each fuel assembly (about 80% of it) with MOX fuel rods and the periphery uranium oxide. In the MOX part, minor actinides are burned as well as recycled plutonium. Power is quoted at 1346 MWe. Little has been heard of the project for several years.

The main rationale for RMWRs is extending the world's uranium resource and providing a bridge to widespread use of fast neutron reactors. Recycled plutonium should be used preferentially in RMWRs rather than as MOX in conventional LWRs, and multiple recycling of plutonium is possible. JAERI started the research on RMWRs in 1997 and then collaborated in the conceptual design study with the Japan Atomic Power Company (JAPCO) in 1998. Hitachi has also been closely involved.

A new reprocessing technology is part of the RMWR concept. This is the fluoride volatility process, developed in 1980s, and is coupled with solvent extraction for plutonium to give Hitachi's Fluorex process. In this, 90-92% of the uranium in the used fuel is volatalised as UF6, then purified for enrichment or storage. The residual is put through a Purex circuit which separates fission products and minor actinides as HLW, leaving the unseparated U-Pu mix (about 4:1) to be made into MOX fuel.

Fast neutron reactors

The 2014 Strategic Energy Plan sets R&D on fast neutron reactors (FNRs) as a priority, including that on waste volume reduction and toxicity decrease.

Among several critical assemblies, JAEA has operated the Fast Critical Assembly (FCA) at Tokai since 1967 to study the neutronic characteristics of fast reactors. It has been used for both Joyo and Monju development. In the course of this the facility has accumulated some hundreds of kilograms of separated plutonium

(reported as being weapons-grade) and high-enriched uranium. In March 2014 it was agreed to send all of this to the USA for disposal or downblending and civil use respectively under the auspices of the Global Threat Reduction Initiative (GTRI) set up by the USA in 2004. The FCA itself will be converted to use low-enriched uranium and reassigned to transmutation and disposition of wastes.

The **Joyo** experimental fast breeder reactor at Oarai operated from 1977 with a succession of three cores, was boosted to 140 MWt in 2003, and has accumulated a lot of technical data. It has been shut down since 2007 due to damage to some core components. The upper core structure had to be replaced, and this was completed in 2014. After substantial upgrading, in 2016 JAEA sought permission from the NRA to restart it in 2021. It intends to complete engineering work on it by the end of 2020, and expects to operate it at 100 MWt from mid-2022 with its Mk IV core comprising up to 79 MOX fuel assemblies.

JAEA said Joyo had irradiated around 100 MOX fuel assemblies during about 71,000 hours of operation, and was significant in Japan's fuel cycle policy. Later it said that 342 fast reactor fuel assemblies had been irradiated over 1982-2000 at 100 MWt, and then 130 at 140 MWt over 2003-2007.

The 280 MWe **Monju** prototype FBR reactor started up in April 1994 and was connected to the grid in August 1995, but a sodium leakage in its secondary heat transfer system during performance tests in December 1995 meant that it was shut down after only 205 days of actual operation.

In mid-2012, the Education, Science & Technology Ministry, MEXT, outlined to the AEC some options for the future of Monju, for which it is responsible through JAEA. If Japan opts for direct underground disposal of used fuel, Monju would be terminated. If the closed fuel cycle with reprocessing is continued, Monju would continue with its original mission to prepare for commercial use of FBRs from 2050, with a demonstration unit to operate from 2025. Monju is reported to have cost JPY 1 trillion ($12.5 billion) to build and operate, and its budget for 2012 was JPY 17.5 billion. Early in 2014 its maintenance was reported as costing JPY 50 million per day.

Monju is discussed in the Fast neutron reactors section of the information paper on *Nuclear Power in Japan*. It has been shut down for most of its life, and in December 2016 the government confirmed that it would be decommissioned.

In May 2014 Japan committed to support the development of the French **Astrid** fast reactor project, and in August 2014 JAEA,

Mitsubishi Heavy Industries and Mitsubishi FBR Systems concluded an agreement with the French Atomic Energy Commission (CEA) and Areva to progress cooperation on Astrid. Astrid was envisaged as a 600 MWe prototype of a commercial series of 1500 MWe sodium-cooled fast reactors which were planned to be deployed from about 2050 to utilise the abundant depleted uranium available by then and also burn the plutonium in used MOX fuel. Astrid will have high fuel burnup, including minor actinides in the fuel elements, and great flexibility in breeding ratios. Astrid is called a 'self-generating' fast reactor rather than a breeder in order to demonstrate low net plutonium production. It arises from a 2006 French government commission to the CEA to develop a fast neutron reactor which is essentially a Generation IV version of the sodium-cooled type which already has 45 reactor-years' operational experience in France. In mid-2009 Astrid (Advanced Sodium Technological Reactor for Industrial Demonstration), was given high priority in R&D on account of its actinide-burning potential. The CEA sought partnerships with Japan and China to develop it.

The Astrid programme includes development of the reactor itself and associated fuel cycle facilities: a dedicated MOX fuel fabrication line (AFC) and a pilot reprocessing plant for used Astrid fuel (ATC) about 2023. Fuel rods containing actinides for transmutation are scheduled to be produced from 2023, though fuel containing minor actinides would not be loaded for transmutation in Astrid before 2025. In 2014, in line with their October 2010 agreement on fast reactor R&D, France had asked Japan to test fuel for Astrid in the Monju fast reactor.

In June 2018 the French government stated that Astrid will have its capacity scaled down from the initially planned 600 MWe to between 100 and 200 MWe to reduce construction costs and also due to development of a commercial fast reactor no longer being a high priority. Following the decision, Toshiba said that the smaller Astrid would be a step back for Japan's fast reactor development process, possibly forcing the country to build its own larger demonstration reactor in Japan rather than rely on Astrid.

Thorium and MSRs

Japan's Research Institute for Applied Sciences based in Kyoto with Chubu Electric at Hamaoka are developing thorium-fueled molten salt reactor technology, from 2013.

Regulation and safety

The **Atomic Energy Commission (JAEC)** was set up in 1956 under the Science & Technology Agency (STA) but has been part of the Cabinet Office since 2001. It was a senior policy body except for a period in 2011 to 2012, though in 2001 it was reduced to an advisory role. A government panel in 2013 recommended that it become simply a committee focused on such issues as the peaceful uses of nuclear energy, nuclear non-proliferation, and the treatment and disposal of radioactive waste. The panel said that it should no longer be involved with formulating nuclear policy or estimating funding requirements, but should remain within the Cabinet Office. In 2014 parliament reduced its role to the supervision of Japan's plutonium and advising on radioactive wastes.

The former Nuclear & Industrial Safety Agency (NISA) within the Ministry of Economy Trade & Industry (METI, the successor of MITI) was responsible for nuclear power regulation, licensing and safety. It conducted regular inspections of safety-related aspects of all power plants.

In mid-2011, following the Fukushima accident, the government established a new and more independent **Nuclear Regulation Authority (NRA)** under the Environment Ministry. This commenced in September 2012 and combined the roles of NISA and NSC, and also the monitoring functions of the Education & Science Ministry (MEXT). The four commissioners and chairman were appointed in February 2013. It started with a staff of 473, nearly three quarters of whom were from NISA, and a budget of ¥50 billion/yr (about $600 million). It is modelled on the US Nuclear Regulatory Commission.

The first task of the NRA was decide on reactor restarts (see section in the information paper on Nuclear Power in Japan). In July 2013 the NRA published two sets of regulations with regard to detailed design of nuclear power plant systems and severe accident management procedures. Following an IAEA Integrated Regulatory Review Service (IRRS) mission to Japan in 2016 the NRA revised its prescriptive checklist approach to reactor inspections and is "fully acting on the IAEA recommendations," with

bills to amend reactor and fuel facility inspection reform laws being submitted to the government in February 2017. The proposed new reactor and fuel law stipulates that the revised inspection system "will come into force within three years after promulgation." This is intended to provide both the companies and the NRA with ample time to prepare.

Regulations relating to the fuel cycle and research reactors including Monju took effect in mid-December 2013. The new regulations apply to two used fuel reprocessing plants – JNFL's Rokkasho commercial plant and the older JAEA facility at Tokai (shut down since 2007); seven fuel fabrication facilities, including JNFL's partially completed mixed-oxide fuel facility; Recyclable-Fuel Storage Co.'s partially completed interim spent fuel storage facility at Mutsu; four waste storage facilities; 22 research reactors; and 15 large and 196 small facilities using nuclear fuel for research. They require facility operators to assume stricter earthquake and tsunami standards in line with new safety guidelines for nuclear reactors implemented in July, and they require improved safety measures to prevent hydrogen explosions and criticalities during emergencies.

Also coming under the remit of the Ministry of the Environment is the new five-member **Nuclear Safety Investigation Commission** (NSIC), which replaced the Nuclear Safety Commission (NSC) – a senior government body set up in 1978 under the Atomic Energy Basic Law and responsible for formulating policy, alongside the Atomic Energy Commission (JAEC). NSIC will review the effectiveness of the NRA and be responsible for the investigation of nuclear accidents. The Environment Ministry already handles disposal of radiation-contaminated debris around the Fukushima Daiichi nuclear plant. The lower house of parliament (Diet) passed the enabling legislation in mid June 2012, with the support of all three main parties, and the upper house endorsed it a week later. The reform was implemented in September. Key issues will be addressed toward creating a stronger and more effective safety regulatory organization, with a plan to be issued by year end. As an expression of its determination to strengthen nuclear safety regulation Japan plans to receive an IAEA Integrated Regulatory Review Service mission later in 2012.

The Japan Nuclear Energy Safety Organisation was set up in 2003 to inspect nuclear installations and nuclear reactor facilities and undertake safety analysis and evaluations of the design of nuclear installations and nuclear reactor facilities. It had 401 staff (as of October 2013, compared with 527 in NRA then) and functioned as technical support for NRA. The law that established NRA in September 2012 stipulated that JNES must be integrated into NRA,

and in October 2013 NRA outlined provisions for the merger by March 2014, when 384 staff transferred. As well as the staff from both organisations, 81 will be added to boost capacity for inspections, making a total of over 1000. The merger substantially enhances the professional competence and experience base of NRA.

In December 2013 the NRA with its China and South Korean counterparts agreed to form a network to cooperate on nuclear safety and quickly exchange information in nuclear emergencies. In addition to exchanging information on nuclear accidents, the three countries will share standard information such as safety plans.

Following the Fukushima accident, the **Energy and Environment Council Council** (Enecan or EEC) was set up by the cabinet office in mid 2011 as the energy arm of the National Policy Unit, being chaired by that minister. The Atomic Energy Commission (JAEC) and Central Environment Council apparently came under Enecan until it was abolished by the new government at the end of 2012.

The Science & Technology Agency was responsible for safety of test and research reactors, nuclear fuel facilities and radioactive waste management, as well as R&D, but its functions were taken over by NISA in 2001.

In June 2012 parliament amended the 1955 Atomic Energy Basic Law to stipulate that nuclear plant operators must prevent the release of radioactive materials at abnormal levels following severe accidents, and that the NRA is to formulate regulations to achieve this.

A new inspection system of nuclear facilities came into effect in 2009, following deliberations on the matter since November 2005. Under the new system, the number of nuclear power plants approved for operation over 40 years was expected to increase, starting with Tsuruga 1.

The Atomic Energy Society of Japan (AESJ) has a Committee for Investigation of Nuclear Safety.

Nuclear safeguards remained with METI after the regulatory functions were removed.

Regulatory history

Well before the Fukushima accident, public support for nuclear power in Japan had been eroded since the 1990s due to a series of

accidents and scandals. The accidents to 2011 were the 1996 sodium leak at the Monju FBR, a fire at the JNC waste bituminisation facility connected with its reprocessing plant at Tokai, and the 1999 criticality accident at a small fuel fabrication plant at Tokai. The criticality accident, which claimed two lives, happened as a result of workers following an unauthorised procedures manual. None of these accidents were in mainstream civil nuclear fuel cycle facilities. However, the 1999 accident did lead to safety improvements at nuclear power plants, notably the establishment of emergency off-site facilities (EOF) at all of them.

Following the 1999 Tokai criticality accident, electric power companies, along with enterprises involved with the nuclear industry established the Nuclear Safety Network (NSnet). The network's main activities were to enhance the safety culture of the nuclear industry, conduct peer reviews, and disseminate information about nuclear safety. In 2005 this was incorporated into the Japan Nuclear Technology Institute (JANTI), as the Safety Culture Division. Peer reviews 'tailored to the corporate structure' are implemented periodically for members of NS net involve in the nuclear fuel cycle of Japan. JANTI's Operating Experience Analysis Division collects and analyses operating experience information that was previously handled by the Central Research Institute of Electric Power Industry (CRIEPI) Nuclear Information Center. The Safety Culture Division cooperates with US Institute of Nuclear Power Operations (INPO) and WANO.

Japan's former Nuclear Safety Commission (NSC) confirmed in April 2002 that using mixed oxide (MOX) fuel is safe, and that its use at up to 18 reactors by 2010 was supported. Senior members of the government have reaffirmed that the country's use of MOX "must happen", and that the government will initiate educational and information programs to win wider acceptance for it.

In 2002 a scandal erupted over the documentation of equipment inspections at Tepco's reactors, and extended to other plants. While the issues were not safety-related, the industry's reputation was sullied. Inspection of the shrouds and pumps around the core is the responsibility of the company, which in this case had contracted it out. In May 2002 questions emerged about data falsification and the significance of cracks in reactor shrouds (used to direct water flow in BWRs) and whether faults were reported to senior management. By May 2003 Tepco had shut down all its 17 reactors for inspections, and by the end of 2003 only seven had been restarted. Replacement power cost on average over 50% more than the 5.9 yen/kWh (5.5 cents US) nuclear generation cost. Tepco

eventually had all its reactors back on line, with the whole fiasco costing it about JPY 200 billion (US$ 1.9 billion).

In 2007 NISA ordered reactor owners to check their records for incidents which should have been reported at the time but were not. This revealed a brief (15 minute) criticality incident during refuelling at Hokuriku's Shika 1 BWR in 1999. A series of deficiencies and errors contributed to the incident, and clearly more should have been learned from it to benefit other operators of boiling water reactors such as Chubu and Tohoku, which have also had control rod anomalies over the last 20 years. Tepco said that its Fukishima I-3 BWR may have experienced criticality over seven hours during an outage in 1978, when control rods slipped out of position. NISA ordered the Shika-1 reactor to be shut down for detailed checks.

Seismic concerns

Because of the frequency and magnitude of earthquakes in Japan, particular attention is paid to seismic issues in the siting, design and construction of nuclear power plants. In May 2007 revised seismic criteria were announced which increased the design basis criteria by a factor of about 1.5 and required utilities to undertake some reinforcement of older plants. See also paper on Nuclear Power Plants & Earthquakes.

In July 2007 the Niigata Chuetsu-Oki earthquake occurred on a fault very close to the Kashiwazaki-Kariwa nuclear power plant, and the ground acceleration exceeded the design parameters for the plant, i.e. it was more severe than the plant was required to be designed for. The operating reactors shut down safely and there was no damage to the main parts of the plant. The government (METI) then set up a 20-member Chuetsu Investigation and Countermeasures Committee to investigate the specific impact of this earthquake on the power station, and in the light of this to identify what government and utilities must address to ensure nuclear plant safety. It acknowledged that the government was responsible for approving construction of the first units in the 1970s very close to what is now perceived to be a geological fault line. It was also agreed that the International Atomic Energy Agency (IAEA) would join Japan's Nuclear Safety Commission in a review of the situation. The second IAEA team confirmed after inspecting key internal components that there was apparently "no significant damage to the integrity of the plant". Ground subsidence damaged much equipment around the seven reactors, but the main part of

each plant is built on bedrock, which had entailed excavation in some places to 45 metres.

In October 2008 NISA presented to the NSC its evaluation of Tepco's report on Kashiwazaki Kariwa, assessing it as "appropriate". It contained the results of Tepco's inspections and assessments of equipment, buildings and other structures at the plant following the July 2007 earthquake. In 2009 the NSC endorsed NISA's recommendation that units 6 & 7 be restarted.

Tsunamis are also a feature of Japan and Kuril Islands. Since 1498 there have been 16 tsunamis with maximum amplitudes above 10 metres (some much more), these having arisen from earthquakes of magnitude 7.4 to 9.0, on average one every 30 years. The accident arising from the 11 March 2011 tsunami is described in the paper on the Fukushima Accident.

International outlook and exports

Apart from some active interest in uranium exploration and mine investment in Australia and Canada to help secure fuel supplies, for many years the Japanese nuclear industry was focused domestically, but in the 1990s it started to look at export possibilities and international collaboration.

Companies such as Hitachi, Mitsubishi Heavy Industries (MHI) and Toshiba formed important alliances internationally or took over major foreign nuclear companies. Their export initiatives are in collaboration with an operator such as Tepco and the Japan Bank for International Cooperation.

In heavy manufacturing, particularly of large forgings, Japan Steel Works (JSW) is generally regarded as the main company and world leader. Other enterprises are also active in export of major reactor components. JSW spent JPY 40 billion ($330 million) from 2007 to increase capacity in advance of orders expected from both China and the USA. It has production and research bases in Hiroshima, Yokohama and Muroran. The Muroran centre, in Hokkaido, hosts the heavy steel works and research laboratory relevant to power generation. Muroran manufactures reactor pressure vessels, steam generator components, generator & turbine rotor shafts, clad steel plates and turbine casings for nuclear power plants. JSW has been manufacturing forgings for nuclear plant components to US Nuclear Regulatory Commission standards since 1974, and around 130 JSW reactor pressure vessels are used around the world – more than one-third of the total.

See also the information paper on Heavy Manufacturing of Power Plants.

At the government level, there were agreements with several governments including Kazakhstan. Then NISA set up the International Nuclear Power Safety Working Group in 2008 to cooperate in the field of nuclear safety with emerging countries, primarily in Asia, planning to introduce and expand their use of nuclear power.

This led in 2009 to an industry-based group, the JAIF International Cooperation Center (JICC), established with government backing to support countries planning to introduce nuclear power generation, and the International Nuclear Energy Cooperation Council, a forum for the relevant Japanese government authorities and private institutes to collaborate in international aid.

In October 2010 industry and government set up the International Nuclear Energy Development of Japan Co Ltd (JINED) to export nuclear goods and services collaboratively. The new company will solicit orders for nuclear power plants from countries such as Vietnam starting their own nuclear power programs, and advise on project and infrastructure development, bolstered by legislative and financing support from the Japanese Government. A separate company will be set up to act as engineering, procurement and construction (EPC) contractor. JINED is associated with JAIF and JICC, and is owned by the government (METI, through Innovation Network Corporation), nine utilities (Chubu, Kansai and Tepco being main shareholders), and three manufacturers (MHI, Toshiba and Hitachi).

For Vietnam's second nuclear power plant, Japan Atomic Power Co. and JINED in 2010 signed an agreement with Electricity of Vietnam (EVN) to undertake the Ninh Thuan II project. The government has appointed Mitsubishi to prepare a PWR proposal and Hitachi to prepare one using BWR technology. The project is seen as a METI initiative.

However, following the Fukushima accident reactor manufacturers are reported to be finding it difficult to do business abroad because Tepco, the main operator partner, is preoccupied with remediation at Fukushima. Establishment of its internal entity, the Fukushima Daiichi Decontamination & Decommissioning Engineering Company, to focus on measures for decommissioning units 1-6 and dealing with contaminated water, may ease this.

The only significant export deal in recent years is MHI's involvement with Areva in contracting to build the Sinop nuclear plant in Turkey.

Itochu is also involved. Here, GdF Suez is to be the operating partner. The deal is under a government-level nuclear energy cooperation agreement. In April 2014 the House of Representatives approved this agreement and the export of Japanese nuclear power plants to Turkey and also to the United Arab Emirates. The approval was backed by the ruling Liberal Democratic Party and junior partner New Komeito, as well as the opposition Democratic Party of Japan.

Another initiative which will lead to exports is Hitachi's purchase of Horizon Nuclear Power PLC in the UK, which will lead to construction of four to six ABWR units at Wylfa and Oldbury. Hitachi plans to sell its interest in Horizon once the plants are operating.

In June 2008 an agreement on high-temperature gas-cooled reactor research was initialled by JAEA and the Kazakhstan Atomic Energy Committee, focused on small cogeneration plants.

Nuclear liability

Japan has not been party to any international liability convention but its law generally conforms to them. Two laws governing them have been revised about every ten years: the Law on Compensation for Nuclear Damage and Law on Contract for Liability Insurance for Nuclear Damage.

Plant operator liability is exclusive and absolute, and power plant operators must provide a financial security amount of JPY 120 billion (US$ 1.4 billion). The government may relieve the operator of liability if it determines that damage results from "a grave natural disaster of an exceptional character", though it did not do this after the Fukushima accident in 2011, and in any case liability is unlimited. For the Fukushima accident the government set up a new state-backed institution to expedite payments to those affected.

In line with a 2013 undertaking by the Minister for Foreign Affairs and confirmed in the 4th Basic Energy Plan adopted in April 2014, a bill to ratify the IAEA's Convention on Supplementary Compensation for Nuclear Damage (CSC) was passed by both houses of parliament in November 2014, along with amended domestic compensation laws. The Ministry of Foreign Affairs has now filed a formal document with the IAEA so that Japan became the sixth member country of the CSC, enabling it to enter force globally in mid-April 2015.

Non-proliferation

The Atomic Energy Basic law prohibits the military use of nuclear energy and successive governments have articulated principles reinforcing this. In 1976 Japan became a party to the Nuclear Non-Proliferation Treaty with its safeguards arrangements administered by the UN's International Atomic Energy Agency, and in 1999 it was one of the first countries to ratify the Additional Protocol with IAEA, accepting intrusive inspections.

Japan is noteworthy in being the only non-weapons state under the NPT with major fuel cycle facilities, which are thus under full safeguards. The Rokkasho reprocessing plant is the first such plant to be under full IAEA safeguards (others are under Euratom safeguards). Monitoring equipment funded by IAEA was built in to the plant, which was a novel challenge for both IAEA and JNFL.

Japan also has bilateral safeguards arrangements with its major nuclear supplier states and has long been a member of the Nuclear Suppliers Group which restricts export of nuclear equipment.

Notes & references

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